Sahaj Bhadja

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Summary

Software engineer pursuing an MSCS with 2+ years building Python/C++ systems, ML evaluation pipelines, containerized services, and distributed workflows across Linux-oriented, API-driven, research-heavy environments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built **ML evaluation pipelines** across 100,000+ records, producing structured metrics, logs, and reports for reproducible experiments, anomaly reviews, and model behavior analysis at scale for researchers.
- Documented **evaluation and reporting workflows** for researchers, turning ad hoc experiments into repeatable processes with clearer validation outputs, debugging notes, and technical reviews across teams during reviews.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered **cross-platform integration services** using Python, SQL, AWS Glue, and S3, improving response time by **30%** while supporting distributed APIs, data movement, and reporting workflows reliably.
- Maintained **20+ automated validation suites** with Jenkins, PyTest, and Mockito, increasing CI reliability by **68%** and reducing regressions across Windows, Linux, and macOS environments consistently for releases.

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built **Azure SQL-backed backend services** for 500+ users, translating product requirements into APIs, validation logic, schema updates, and production releases across cloud-hosted data workflows and systems reliably.
- Improved **service reliability and performance** by tuning indexed queries, background jobs, and CI/CD releases, reducing dashboard latency from **1.2s to 350ms** while supporting operational debugging and handoffs daily.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built **REST and WebSocket modules** over SQL/NoSQL persistence, supporting 2M+ synchronized records across integrations, dashboards, and operational workflows for high-volume users and systems during daily internal operations.
- Automated **Docker-based build and test flows** with Jenkins, cutting release time by **50%** while improving deployment consistency, troubleshooting, and team handoffs during releases and validations weekly.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed **Python services and REST APIs** handling 10,000+ daily requests, integrating uploads, OCR processing, persistence workflows, and automated checks for production reliability under active usage daily.
- Debugged **runtime and workflow defects** across upload, OCR, persistence, and grading failures, documenting fixes and reducing feedback turnaround by **60%** for production users and operators consistently.

Skills

Languages: **Python**, **C++**, JavaScript, TypeScript, SQL, Java, Bash, HTML/CSS
ML/AI: PyTorch, TensorFlow familiarity, model evaluation, OCR workflows, multimodal pipelines, structured outputs
Systems: Linux development, REST APIs, WebSockets, JSON/protobuf-style data formats, distributed workflows
Cloud/Infra: Docker, Jenkins, AWS Glue, S3, Azure SQL, PostgreSQL, MySQL, MongoDB, CI/CD
Practices: Debugging, automated validation, code reviews, technical documentation, Agile/Scrum, AI-assisted development

Projects

CaptionCraft Live | *Transformers, PyTorch, Flask, Computer Vision*
- Benchmarked **BLIP, GIT, ViT-GPT-2, and custom vision models** on 10K images and 100K captions, improving caption accuracy from **81%** to **96%** through evaluation pipelines consistently.

CPU Scheduling Simulator Live | *C++, React, JavaScript, OS Algorithms*
- Implemented **FCFS, SJF, SRT, RR, priority, and MLQ** schedulers with C++ logic, debugging workflows, validation, timing analysis, and interactive visualization across simulated workloads reliably for testing exercises.

Big-Brain Live | *LLaMA-2, FAISS, PostgreSQL, GCP*
- Built a **semantic retrieval platform** over 50K+ documents with vector search, summarization, and contextual Q&A workflows supporting secure, enterprise-style document analysis for teams and analysts daily.